

ASSIGNMENT - 5

GOWTHAM PATEL : C
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4) Encrypt the message "ENJOY YOUR JOURNEY" using Rail fence with depth 3 & 2.

i) Depth 2:

E J Y O U R O R E
N O Y U J U N Y

CT = EJYORORE NOYUJUNY

ii) Depth 3:

E Y R R
N O Y U J U N Y
J O O E

CT = EYRR NOYUJUNY JOOE

5) Encrypt the message as per your choice using Rail Fence with depth 2 & 3. PT = GOOD DAY

i) Depth 2:

G O D Y
O D A

CT: GODYODA

ii) Depth 3:

G A D L E S
O D A A J W
O Y

CT: GDODAOY

6) Decrypt the cipher "EOCS NYUWLEJYREASON". Encrypted using Rail Fence with depth = 4

Depth = 4, Length = 19

E				O				C						S
	N			Y		U		W		L			E	
		J		Y		R		E		A		S		
			O				N				S			

PT = ENJOY YOUR NEW CLASSES

4) Decrypt the cipher "~~SEMSERTISINCMOO~~" "BOOI GUYUBOSNUOTYRKAS". Encrypted using Rail fence with depth = 3

∴ Depth = 3 length = 20

B				O				O				I				G			
	U		Y		U		B		O		S		N		U		U		T
		Y				R				K				A					S

PT: BUY YOUR BOOKS IN AUGUST

5) Decrypt the cipher "SEMSERTISINCMOO" encrypted using Rail fence with depth 2.

∴ Depth = 2 Length = 15

S . E . M . S . E . R . T . I
S I N C M O O

PT: SEMSERTISINCMOO

Permutation

1) Encrypt the message using key 5 3 1 2 4.
"WE ARE DISCOVERED".

5 3 1 2 4
W E A R E
D I S C O
V E R E D

CT: A S R R C E E I E E O D W D V

DES: Data Encryption Standards.

1) using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as: 0x002 000 0000 0001

Binary form: 0000 0000 0000 0010
0000 0000 0000 0000
0000 0000 0000 0000
0000 0000 0000 0001

0000 0000 0000 0010 0000 0000 0000 0000
 0000 0000 0000 0000 0000 0000 0010

Input bit 64 = 1 , o/p position 25 = 7

Input bit 18 = 7 o/p position 63 = 1

0000 0000 0000 0000 0000 0000 1000 0000
 0000 0000 0000 0000 0000 0000 0000 0010

o/p in hexadecimal : 0x000 008 0000 0001

2) Given the 4 bit plaintext [M] 1101 & key [K] 1011.
 Calculate the resulting ciphertext [C] using a bitwise exclusive OR
 [XOR] operation.

: PT = 1101
 Key = 1011

 CT = 0110

3) The input to S box is 000000 what is o/p

: 000000
 $\downarrow$ $\downarrow$
 C C

as 0th row & 0th column in S box is 14.

14 $\rightarrow$ 2^3 2^2 2^1 2^0
 1 1 1 0

[o/p = 1110]